

Paper Review

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Title & Author

Title: Multi-Agent Digital Twins for Strategic Decision-Making using Active Inference

Authors: Francesco Maria Mancinelli, Matteo Torzoni, Domenico Maisto, Francesco Donnarumma, Alberto Corigliano, Giovanni Pezzulo, and Andrea Manzoni

Source: arXiv:2604.12657 [cs.CE], April 14, 2026.

Summary

This paper proposes a multi-agent Digital Twin framework in which autonomous agents use Active Inference to update beliefs, select policies, and coordinate through a shared environment. The paper also proposes two extensions to the basic Active Inference framework, a contextual inference technique for dealing with non-stationary environments that handle uncategorized evidence, and a streaming machine learning approach for updating preference-related parameters of the agent over time rather than requiring that all these parameters are known a priori. This new architecture is evaluated through several simulations modeled on the classic theory of Cournot competition in both a two-firm and three-firm setting.

Strengths

Emphasis is placed on the agents' ability to infer global context and act, accordingly, demonstrating the expected behavior that would be optimal if the underlying assumptions of the agents' generative model were accurate. This paper also illustrates how the emerging field of Digital Twins may provide a viable platform for applicability of the proposed extensions.

Major Comments

The paper motivates a compact generative model as a tool for efficient scaling. However, the models clearly rely on a fair amount of hand-design. It is possible that the proposed matrices are too basic for the Cournot setting. Are there many more possibilities for likelihood, transition, and preference matrices such as hidden states of the market, different actions (advertising), different observation modalities (competitor actions), and context variables (weather) are possible. A sensitivity analysis would be especially useful if it is easy to do and if it identifies the most problematic cases.

Also, the idea of scaling up many agents is only investigated weakly. Three firms are not very large. In theory, the approach could allow for an arbitrary number of firms, but what about the expected free-energy policy evaluation? How does it scale up with the number of state factors, action factors, agents, and economic horizon? It would be best to have an asymptotic analysis in terms of computational costs or runtime experiments.

I would also be interested in seeing the performance compared to. For example, reinforcement learning, model predictive control, or even simple online forecasting policies with the same information constraints. In particular, the conclusion that the policies are remarkably robust and perform well on unseen data seems to be questionable. Probably, it also depends on the structure of the generative model, which is not further investigated, and that also plays an important role.

Finally, the augmented-precision experiments presented are probably interesting but are too strongly based on visual assessment of the supplementary plots. For the augmented-precision parameter, some kind of plot with means over random seeds would help. Also, reporting cumulative profit, regret relative to Nash equilibrium, inventory cost, prediction error, and policy entropy would be very useful.

Minor Comments

- Several multi-panel figures are dense; larger legends and higher-contrast palettes would improve readability, especially in grayscale.
- Key acronyms, including GM, GP, VFE, EFE, and SML, should be redefined at first use in major sections.

APA Reference

Mancinelli, F. M., Torzoni, M., Maisto, D., Donnarumma, F., Corigliano, A., Pezzulo, G., & Manzoni, A. (2026). *Multi-agent digital twins for strategic decision-making using active inference*. arXiv. <https://doi.org/10.48550/arXiv.2604.12657>